

JAHANVI RANA

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LinkedIn — Portfolio — GitHub — Kaggle

Professional Summary

Data Analyst and Business Intelligence enthusiast skilled in Excel, SQL, Python, and Power BI, with hands-on experience building end-to-end analytics solutions, interactive KPI dashboards, and predictive analytics models. Co-authored an international Computer Vision research publication and experienced in transforming raw data into actionable business insights through data cleaning, ETL, visualization, statistical analysis, and business reporting.

Core Competencies

Business Intelligence · Data Visualization · Predictive Analytics · Dashboard Development · Machine Learning · Data Cleaning · ETL · Data Modeling · KPI Reporting · Forecasting · Statistical Analysis · Decision Support · Data Pipeline · Analytical Problem Solving · Communication

Technical Skills

Programming & Libraries: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), SQL

Data Analytics: Exploratory Data Analysis (EDA), Data Cleaning, Validation, Feature Engineering, Time-Series Analysis, Data Storytelling, Data Pipeline

Business Intelligence: Power BI (Power Query, DAX, Data Modeling, KPI Dashboards, Drill-through, Bookmarks)

SQL Competencies: Joins, Aggregate Functions, Subqueries, Views, Filtering, Grouping

Excel & Productivity Tools: Pivot Tables/Charts, Power Query, Conditional Formatting, VLOOKUP, Git, GitHub, Google Colab, Kaggle, Jupyter, Word, PowerPoint, Overleaf

Projects

1.) OTT Streaming Analytics Dashboard

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Tools: Python · SQL · Power BI · DAX · Excel · Pandas · Matplotlib

Impact: Designed an end-to-end BI solution analyzing engagement, growth, and revenue trends for an OTT platform.

- Built automated Python (Pandas) pipelines for data cleaning, transformation, validation, and exploratory analytics.
- Designed an interactive multi-page Power BI dashboard detailing Executive KPIs, Segmentation, Geography, and Content.
- Developed dynamic DAX measures for core business metrics (Monthly Revenue, Active Users, Retention, Conversion).
- Integrated drill-throughs, bookmarks, and slicers to deliver interactive navigation and decision support.

2.) Gold Market Intelligence & Predictive Forecasting System

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Tools: Python · Power BI · XGBoost · LightGBM · CatBoost

Impact: Built a predictive analytics platform forecasting global and Indian gold prices by merging macro indicators with ML.

- Collected, cleaned, and transformed multi-source financial datasets while engineering complex time-series features.
- Developed and compared XGBoost, LightGBM, and CatBoost ensemble models using MAE, RMSE, and R^2 metrics.
- Built Power BI dashboards visualizing Price Trends, Macro Economic Indicators, Heatmaps, and Forecasting Analysis.

Research Publication

Research Co-Author

2025

“Enhancing YOLO with Rotated Bounding Boxes and ResNet50 Backbone for Accurate Orientation-Aware Object Detection,” *Proceedings of the International Conference on Innovative Computing and Communication (ICICC 2026)*, hosted on SSRN. Achieved a 13.5% increase in Mean Average Precision (mAP) and an 18% improvement in rotated Intersection over Union (IoU) on the DOTA dataset using a custom circular IoU loss function. **View Publication**

Education

Chandigarh University

Mohali, Punjab

Bachelor of Engineering (Computer Science – Artificial Intelligence & Machine Learning)

2022 – 2026

CGPA: 7.75/10

M M M Public School

Hoshiarpur, Punjab — 2022

Intermediate (Non-Medical with Computer Science)

Percentage: 88.8%

SGGS Sr. Sec Public School

Mukerian, Punjab — 2020

High School / Matriculation

Percentage: 75%

Training & Certifications

• **Python Essentials 1 & 2** – Cisco Networking Academy & OpenEDG Python Institute

Verify — Jun 2023

• **Preparing Data for Analysis with Microsoft Excel** – Microsoft Learn

Verify — Jul 2025

• **Extract, Transform and Load Data in Power BI** – Microsoft Learn

Verify — Sep 2025

• **Data Modeling in Power BI** – Microsoft Learn

Verify — Jan 2026